Problem Set 7 - Solutions

Problem 4.22

What is the standard heat of combustion of n-pentane gas at 25°C if the combustion products are H_2O(l) and CO_2(g)?

Solution

Balanced Reaction:

$$C_5H_{12}\,(g) + 8\,O_2\,(g) \rightarrow 5\,CO_2\,(g) + 6\,H_2O\,(l)$$

In[ ]:=
```
ν = {-1, -8, 5, 6};
ΔHf298 = {-146 760, 0, -393 509, -285 830};(*J/mol*)
ΔH298 = ν.ΔHf298
```

Out[ ]=

$-3\,535\,765$

```
(*J per mole n-pentane*)
```

The standard heat of combustion of n-pentane gas at 25 °C with products H_2O(l) and CO_2(g) is -3,535,765 J per mol of n-pentane. //ANS

Problem 4.28

Natural gas (assume pure methane) is delivered to a city via pipeline at a volumetric rate of 150 million standard cubic feet per day. If the selling price of the gas is $5.00 per GJ of higher heating value, what is the expected revenue in dollars per day? Standard conditions are 60°F and 1 atm.

Solution

Balanced combustion reaction:

$$CH_4\,(g) + 2\,O_2\,(g) \rightarrow CO_2\,(g) + 2\,H_2O\,(liq)$$

In[ ]:=
```
ν = {-1, -2, 1, 2};
ΔHf298 = {-74 520, 0, -393 509, -285 830};(*J/mol*)
ΔH298 = ν.ΔHf298(*HHV*)
```

Out[ ]=

$-890\,649$

In[ ]:=
```
(*n=P*V/R*T*)
R = 0.7302 * ft^3 atm / (lbmol * degR);

T = (60 + 459.67) * degR;

n = (150 000 000 * ft^3/d * 1 atm) / (R * T) * (1 000 gmol) / (2.20462 lbmol)
```

Out[ ]=

$$\frac{1.793\,032 \times 10^8\ \text{gmol}}{\text{d}}$$

Convert gram-moles to GJ and the GJ to dollars:

In[ ]:=
```
n * Abs[ΔH298] J / gmol * 1 GJ / 10^9 J * 5 dollars / GJ
```

Out[ ]=

$$\frac{798\,481.1\ \text{dollars}}{\text{d}}$$

The expected revenue is $798,481.10 per day. //ANS

Problem 4.53

Saturated water vapor, i.e., *steam*, is commonly used as a heat source in heat exchanger applications. Why *saturated* water vapor? Why saturated *water* vapor?

In a plant of any reasonable size, several varieties of saturated steam are commonly available. For example, saturated steam may be available at 4.5, 9, 17, and 33 bar. But the higher the pressure, the lower the useful energy content and the greater the cost. Why is the energy content lower at higher pressure? Why then is higher-pressure steam used?

Solution

Saturated vapor is used because latent heat ΔH_{lv} is large compared to sensible heat. As a result, latent heat will overwhelm sensible heat in enthalpy calculations. //ANS

Water is used because it is cheap, readily available, non-toxic, and has a large ΔH_{lv}. //ANS

The "energy content" of (saturated) steam is lower at higher pressures because the molar volume decreases as pressure is increased . This can be seen in the ideal gas equation PV=RT. From the Clapeyron equation, with $\Delta V \sim V$ and noting that dP^{sat}/dt is unaffected by pressure, we see that ΔH_{lv} is "roughly" proportional to V. The lower energy content at higher pressure is a result of the decrease in ΔH_{lv} with increasing P and decreasing V. //ANS

Out[ ]//TraditionalForm=

$$\Delta H_{lv} = T\,\Delta V \frac{dP^{sat}}{dT}$$

Out[•]= Traditional form

$$\Delta H_{lv} \approx T\, V\, \text{const}$$

This conclusion can be verified with the data in the steam tables. A few values are shown below from Table E.1 in Appendix E. As pressure is increased, vapor volume decreases, while T and liquid volume increase. However, those increases are not enough to compensate for the for the relatively large decrease in volume.

T	T	P	V		ΔV	T·ΔV	dPsat/dT	T·ΔV·(dPsat/dT)
°C	K	kPa	cm^3/g		cm^3/g	$K \cdot cm^3/g$	kPa/K	$kPa \cdot cm^3/g$
			sat. liq.	sat .vap.				
80	353.15	47.36	1.029	3409.1	3408.1	1,203,560	1.916	2,306,021
90	363.15	70.11	1.036	2361.3	2360.3	857,130	2.659	2,279,108
100	373.15	101.33	1.044	1673.0	1672.0	623,890	3.611	2,252,868
110	383.15	143.27	1.052	1209.9	1208.8	463,170	4.809	2,227,385
120	393.15	198.54	1.061	891.5	890.4	350,076	6.289	2,201,629
150	423.15	476.00	1.091	392.4	391.3	165,582	12.808	2,120,779

High-pressure steam is used because higher pressures allow higher temperatures. //ANS

Problem 4.54

The oxidation of glucose provides the principal source of energy for animal cells. Assuming the reactants are glucose [$C_6 H_{12} O_6(s)$] and oxygen [$O_2(g)$] and the products are $CO_2(g)$ and $H_2 O(l)$, answer the following:

(a) Write a balanced equation for glucose oxidation, and determine the standard heat of reaction at 298 K.

(b) During a day, an average person consumes about 150 kJ of energy per kg of body mass. Assuming glucose is the only source of energy, estimate the mass (grams) of glucose required daily to sustain a person of 57 kg.

(c) For a population of 275 million people, what mass of CO_2 (a greenhouse gas) is produced daily by mere respiration? Data: for glucose, $\Delta H^o_{f,298}$ = -1,274.4 kJ/mol. Ignore the effect of temperature on the heat of reaction.

Solution - Part (a)

Balanced oxidation (combustion) reaction:

$$C_6 H_{12} O_6(s) + 6\, O_2\,(g) \rightarrow 6\, CO_2\,(g) + 6\, H_2O\,(liq) \quad \text{//ANS}$$

Standard heat of reaction at 298K:

In[•]:=
```
ν = {-1, -6, 6, 6};
ΔHf298 = {-1274.4, 0, -393509 / 1000, -285830 / 1000};
ΔHrxn = ν.ΔHf298
```

Out[•]=
```
-2801.634
```

The standard heat of reaction at 298 K is -2,801.63 kJ per mole of glucose. //ANS

Solution - Part (b)

Total energy consumption for a 57-kg person:

In[ ]:=
$$\frac{150\ \text{kJ}}{\text{kg}} * 57\ \text{kg}$$

Out[ ]=
$$8\,550\ \text{kJ}$$

Moles of glucose required:

In[ ]:=
$$\frac{8\,550\ \text{kJ}}{2\,801.63\ \text{kJ}/\text{mol}}$$

Out[ ]=
$$3.051\,795\ \text{mol}$$

Mass required:

In[ ]:=

molar mass of glucose »

D-(+)-glucose CHEMICAL [molar mass]

Out[ ]=
$$180.16\ \text{g}/\text{mol}$$

The molar mass of glucose is 180.16 g/mol.

In[ ]:=
$$\frac{3.05\ \text{mol}}{\text{day}} * \frac{180.16\ \text{g}}{\text{mol}}$$

Out[ ]=
$$\frac{549.488\ \text{g}}{\text{day}}$$

The 57-kg person requires about 550 g of glucose. //ANS

Solution - Part (c)

In[ ]:=

molar mass of CO2

carbon dioxide CHEMICAL [molar mass]

Out[ ]=
$$44.009\ \text{g}/\text{mol}$$

In[ ]:=
$$\frac{3.05\ \text{mol}}{\text{day}} * \frac{6\ \text{mol}}{1\ \text{mol}} * 275\,000\,000 * \frac{44\ \text{g}}{\text{mol}} * \frac{1\ \text{kg}}{1\,000\ \text{g}} * \frac{1\ \text{tonne}}{1\,000\ \text{kg}}$$

Out[ ]=
$$\frac{221\,430.\ \text{tonne}}{\text{day}}$$

221,430 metric tons of CO_2 are produced per day. //ANS

Problem 4.45

A process for the production of 1,3-butadiene results from the catalytic dehydrogenation at atmospheric pressure of 1-butene according to the reaction:

$$C_4H_8\,(g) \rightarrow C_4H_6\,(g) + H_2\,(g)$$

To suppress side reactions, the 1-butene feed is diluted with steam in the ratio of 10 moles of steam per mole of 1-butene. The reaction is carried out *isothermally* at 525°C, and at this temperature 33% of the 1-butene is converted to 1,3-butadiene. How much heat is transferred to the reactor per mole of entering 1-butene?

Solution

Step 1: Balance the Chemical Equation

$$C_4H_8\,(g) + 10\,H_2O\,(g) \rightarrow C_4H_6\,(g) + H_2\,(g) + 10\,H_2O\,(g)$$

Step 2: Determine the Heat of Reaction at 298K

```
(*List order: C4H8, H2O, C4H6, H2, H2O*)
```

In[•]:=
```
ν = {-1, -10, 1, 1, 10};
ΔHf298 = {-540, -241 818, 109 240, 0.000, -241 818};
ΔH298 = ν.ΔHf298 ;
```

Step 3: Determine MDCPH

In[•]:=
```
a = {1.967, 3.470, 2.734, 3.249, 3.470};
b = {31.630, 1.450, 26.786, 0.422, 1.450} * 10^-3;
c = {-9.873, 0.000, -8.882, 0.000, 0.000} * 10^-6;
d = {0.000, .121, 0.000, 0.083, 0.121} * 10^5;
```

In[•]:=
```
Δa = ν.a;
Δb = ν.b;
Δc = ν.c;
Δd = ν.d;
```

In[•]:=
```
T1 = 525 + 273.15;
T0 = 25 + 273.15;
```

In[•]:=
```
R = 8.314; (*J/(mol*K)*)
```

In[•]:=
```
MDCPH = Δa + Δb/2 * (T1 + T0) + Δc/3 * (T1^2 + T0^2 + T1 * T0) + Δd/(T1 * T0);
```

Step 4: Determine the Heat of Reaction at T2

In[•]:=
```
ΔH798 = ΔH298 + MDCPH * R * (T1 - T0)
```

Out[•]=
```
117 866.9
```

```
In[•]:= Q = .33 * ΔH798
Out[•]=
38 896.08
```

The heat requirement is 38,896.1 J per mol 1-butene. //ANS

Alternate Solution Using IDCPH

```
In[•]:= IDCPH = Δa * (T1 - T0) + Δb/2 * (T1^2 - T0^2) + Δc/3 * (T1^3 - T0^3) + Δd * ((T1 - T0)/(T1 * T0));
ΔH798 = ΔH298 + IDCPH * R;
Q = .33 * ΔH798
Out[•]=
38 896.08
```

Alternate Solution Using Integrals

```
In[•]:= ΔCp = Δa + Δb * T + Δc * T^2 + Δd * T^-2;
ΔH798 = ΔH298 + R * ∫_T0^T1 ΔCp dT;
Q = .33 * ΔH798
Out[•]=
38 896.08
```

Problem 4.55

A natural-gas fuel contains 85mol-% methane, 10 mol-% ethane, and 5 mol-% nitrogen.

(a) What is the standard heat of combustion (kJ/mol) of the fuel at 25°C with H_2O(g) as a product?

(b) The fuel is supplied to a furnace with 50% excess air, both entering at 25°C. The products leave at 600°C. If combustion is complete and if no side reactions occur, how much heat (kJ per mole of fuel) is transferred in the furnace?

Solution - Part (a)

Step 1: Balance the Chemical Equation

$$0.85\ CH_4 + 0.10\ C_2H_6 + 3.075\ O_2 + 11.618\ N_2 \rightarrow 1.05\ CO_2 + 2.00\ H_2O + 1.025\ O_2 + 11.618\ N_2$$

Step 2: Determine the Heat of Reaction at 298K

```
(*List order: CH4, C2H6, O2, N2, CO2, H2O, O2, N2*)
In[•]:= ν = {-0.85, -0.10, -3.075, -11.618, 1.05, 2.00, 1.025, 11.618};
ΔHf298 = {-74 520, -83 820, 0.000, 0.000, -393 509, -241 818, 0.000, 0.000};
ΔH298 = ν.ΔHf298
Out[•]=
-825 096.5
```

ΔH298 = -825.0965 kJ per mol of natural gas fuel. //ANS

Solution - Part (b)

Step 3: Determine MDCPH

```
In[ ]:= a = {1.702, 1.131, 3.639, 3.280, 5.457, 3.470, 3.639, 3.280};
        b = {9.081, 19.225, 0.506, 0.593, 1.045, 1.450, 0.506, .593} * 10^-3;
        c = {-2.164, -5.561, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000} * 10^-6;
        d = {0.000, 0.000, -0.227, 0.040, -1.157, 0.121, -.227, 0.040} * 10^5;

In[ ]:= products = 5 ;; 8;(*Reacants in feed are at T0*)
        Δa = ν[[products]].a[[products]];
        Δb = ν[[products]].b[[products]];
        Δc = ν[[products]].c[[products]];
        Δd = ν[[products]].d[[products]];

In[ ]:= T1 = 600 + 273.15;
        T0 = 25 + 273.15;

In[ ]:= R = 8.314; (*J/(mol*K)*)

In[ ]:= MDCPH = Δa + Δb/2 * (T1 + T0) + Δc/3 * (T1^2 + T0^2 + T1 * T0) + Δd/(T1 * T0);
```

Step 4: Determine the Heat of Reaction at T2

```
In[ ]:= ΔH873 = ΔH298 + MDCPH * R * (T1 - T0)
Out[ ]=
        -533952.1
```

Q = -533.9521 kJ per mol of natural gas fuel. //ANS

Alternate Solution Using IDCPH

```
In[ ]:= IDCPH = Δa * (T1 - T0) + Δb/2 * (T1^2 - T0^2) + Δc/3 * (T1^3 - T0^3) + Δd * ((T1 - T0)/(T1 * T0));
        ΔH873 = ΔH298 + IDCPH * R
Out[ ]=
        -533952.1
```

Alternate Solution Using Integrals

```
In[ ]:= ΔCp = Δa + Δb * T + Δc * T^2 + Δd * T^-2;
        ΔH798 = ΔH298 + R * ∫_T0^T1 ΔCp dT
Out[ ]=
        -533952.1
```

Problem 4.6

If the heat capacity of a substance is correctly represented by an equation of the form

$$C_P = A + BT + CT^2,$$

show that the error resulting when $\langle C_P \rangle_H$ (MCPH) is assumed equal to C_P evaluated at the arithmetic mean of the initial and final temperatures is

$$C(T_2 - T_1)^2 / 12.$$

Solution

```
In[ ]:= ClearAll["Global`*"]
```

```
In[ ]:= Cp = A + B * T + C * T^2;
        MCPH = A + B/2 (T2 + T1) + C/3 * (T2^2 + T1^2 + T2 * T1);
```

IF MCPH is assumed equal to C_P evaluated at the arithmetic mean of the initial and final temperatures T_1 and T_2, THEN the "error" is the difference between the two expressions. If the assumption is true, this difference would be zero. But we are told that the error is $C(T_2 - T_1)^2 / 12$. So compare this given error with an expression for the error as the difference between MCPH and C_P evaluated at the arithmetic mean of T_1 and T_2:

```
In[ ]:= expr1 = MCPH - (Cp /. T → (T1 + T2)/2) // Expand
```

```
Out[ ]=
```

$$\frac{C\ T1^2}{12} - \frac{C\ T1\ T2}{6} + \frac{C\ T2^2}{12}$$

And we would like to show that this is equal to $C(T_2 - T_1)^2 / 12$. Set this up as a second expression (given):

```
In[ ]:= expr2 = C/12 * (T1 - T2)^2 // Expand
```

```
Out[ ]=
```

$$\frac{C\ T1^2}{12} - \frac{C\ T1\ T2}{6} + \frac{C\ T2^2}{12}$$

Compare:

```
In[ ]:= expr1 == expr2
```

```
Out[ ]=
```

```
True
```

```
(*expr1==expr2 is a "boolean" equation - answer is true or false*)
```

Comparison of expressions 1 and 2 shows they are equal. //ANS

Alternate Solution 1:

```
In[ ]:= MCPH - (Cp /. T → (T1 + T2)/2) == C/12 * (T1 - T2)^2 // Expand
```

```
Out[ ]=
```

```
True
```

Alternate Solution 2:

In[ ]:=
$$\text{MCPH} - \left(\text{Cp} /. \text{T} \to \frac{\text{T1} + \text{T2}}{2}\right) == \frac{\text{C}}{12} * (\text{T1} - \text{T2})^2 \;//\; \text{Reduce}$$

Out[ ]=

True